



PPO Training Diagnostics Guide

This reference explains the formulas, target ranges, and actionable interpretations for each legend trace in the **Agent Training Diagnostics** chart.

1. Total Loss & Policy Loss

- **Policy Loss** (L_{policy})

$$\mathcal{L}_{CLIP}(\theta) = -\hat{\mathbb{E}}_t \left[\min \left(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right]$$

Where $r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)}$ is the probability ratio and $\hat{A}_t$ is the estimated advantage.

- **Interpretation:** Measures policy update direction.
- **Good Trend:** Fluctuates near small negative or zero values.
- **Bad Trend:** Drops drastically to large negative numbers or explodes upward (indicates policy destabilization).

- **Total Loss** (L_{total})

$$L_{total} = L_{policy} + c_1 \cdot L_{value} - c_2 \cdot H_{entropy}$$

- **Interpretation:** Weighted combination of all sub-losses.
- **Good Trend:** Smooths out and converges over epochs.
- **Bad Trend:** Continuous upward trend or sharp spikes (indicates hyperparameter mismatch).

2. Value Loss & Explained Variance (Critic Performance)

- **Value Loss** (L_{value})

$$\mathcal{L}_{VF}(\phi) = \frac{1}{2} \hat{\mathbb{E}}_t \left[(V_\phi(s_t) - V_t^{target})^2 \right]$$

- **Interpretation:** Mean Squared Error (MSE) of Critic's expected return predictions vs actual discounted rewards.
- **Good Trend:** Decreases steadily and plateaus at a low baseline.
- **Bad Trend:** Increases continuously or oscillates wildly (Critic failing to evaluate market states).

- **Explained Variance** (e_{var})

$$y_{var} = \text{Var}(V^{target}), \quad e_{var} = 1 - \frac{\text{Var}(V^{target} - V_\phi(s))}{\text{Var}(V^{target})}$$

- **Interpretation:** Percentage of return variance explained by Critic ($1.0 = 100\%$ accuracy).
- **Good Trend:** Rises towards $0.50 \rightarrow 1.00$.
- **Bad Trend:** < 0.0 (Critic is performing worse than predicting a simple static mean return).

3. Entropy (Exploration / Curiosity)

- **Entropy (H)**

$$H(\pi_\theta(\cdot|s_t)) = \hat{\mathbb{E}}_t[-\ln \pi_\theta(a_t|s_t)]$$

For Gaussian policy with covariance matrix Σ : $H = \frac{k}{2}(1 + \ln(2\pi)) + \frac{1}{2} \ln |\Sigma|$

- **Interpretation:** Measures policy randomness and continuous exploration.
 - **Good Trend:** Smooth, gradual decay from high initial state down to a steady plateau (transitioning from exploration to exploitation).
 - **Bad Trend:** Drops abruptly to ≈ 0 in early epochs (**Premature Convergence** / Agent gets stuck in sub-optimal strategy) or never drops (Agent fails to learn).
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4. PPO Stability Metrics (Approx KL & Clip Fraction)

- **Approx KL (D_{KL})**

$$D_{KL} \approx \frac{1}{2} \hat{\mathbb{E}}_t \left[(\ln \pi_{\theta_{old}}(a_t|s_t) - \ln \pi_\theta(a_t|s_t))^2 \right]$$

- **Interpretation:** Quantifies how far policy distribution moves in a single update step.
- **Good Range:** $0.005 \leq D_{KL} \leq 0.030$.
- **Bad Trend:** > 0.05 (Updates too large; risk of policy collapse) or ≈ 0.000 (Agent frozen; non-learning).

- **Clip Fraction (Frac_{clip})**

$$\text{Frac}_{clip} = \hat{\mathbb{E}}_t [\mathbb{I}(|r_t(\theta) - 1| > \epsilon)]$$

- **Interpretation:** Percentage of mini-batch updates hitting clipping threshold ϵ (typically 0.2).
 - **Good Range:** $0.05 \leq \text{Frac}_{clip} \leq 0.20$ (5% – 20%).
 - **Bad Trend:** > 0.30 (Learning rate or update epochs too high) or < 0.01 (Learning rate too small).
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5. Average Step Reward

- **Average Step Reward ($\bar{R}$)**

$$\bar{R} = \frac{1}{N} \sum_{t=1}^N R_t$$

Where R_t is the raw or penalized alpha reward received per environment step.

- **Interpretation:** Agent's out-of-sample/in-sample performance signal during training.
 - **Good Trend:** Consistent upward trajectory across training epochs.
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Quick Diagnostic Reference Summary

Diagnostic Metric	Healthy Range / Trend	Primary Remedy if Unhealthy
Value Loss	Low & Steady Decay	Lower LR or increase Critic network capacity
Explained Var	0.50 → 1.00	Increase Value Network depth / Decrease reward noise
Entropy	Gradual Monotonic Decay	Increase/decrease ENT_COEF
Approx KL	0.005 → 0.030	Lower LR or reduce UPDATE_EPOCHS
Clip Fraction	0.05 → 0.20	Adjust PPO clip parameter ϵ or batch size
Avg Reward	Upward Trend	Tweak advantage normalization or reward shaping penalty